

HTML Forms – Today's Learning Summary

A complete overview of HTML form elements, attributes, and best practices

1. What is an HTML Form?

An HTML form is used to **collect user input** and send it to a server for processing. It acts as a container for different types of input fields.

★ Basic Form Syntax

```
<form action="destination.com" method="get"> ... </form>
```

Attribute	Purpose
action	URL where the form data is sent
method	"get" (visible in URL) or "post" (hidden, more secure)

2. ■ Important Input Types (Highlighted)

Input Type	Usage	Example from your code
text	Single-line text entry	Full Name, First Name, Last Name
password	Hidden text (masked)	Password field in registration form
email	Validates email format	Email Address in donation form
number	Numeric input only	Phone Number, Other Amount
radio	Select ONE option from a group	Donation Amount, Payment Mode, Gender
checkbox	Select MULTIPLE options	Purpose (Medical, Education...), Hobbies
file	Upload files from device	Payment Proof, Profile Picture

3. Other Key Form Elements

<textarea> — Multi-line text input. Used for Comments / long messages.

<select> + <option> — Dropdown list. Used for Country selection in both your forms.

<label> — Links text to an input using the **for** attribute matching the input's **id**.

<button> — Submits the form. Place inside the **<form>** tag.

4. ■ Important Attributes Learned

required: Makes a field mandatory — form won't submit if empty.

placeholder: Shows hint text inside the input field.

maxlength: Limits the number of characters a user can type.

value: Pre-sets or defines the data submitted for radio/checkbox.

name: The key sent to the server with the field's value. MUST be unique per field.

id: Links <label for='...'> to the input. Should be unique on the page.

checked: Pre-selects a radio button or checkbox by default.

type: Defines the kind of input (text, radio, checkbox, file, number, etc.).

5. ■■ Common Mistakes to Fix

- **Duplicate IDs:** Multiple radio buttons had `id="Donation Amount"`. Each element must have a **unique id**.
- **Same name for radio group:** Radio buttons in the same group share one **name**, but each needs a unique **id** for label linking.
- **Spaces in name attribute:** `name="contact number"` — avoid spaces; use `name="contact_number"`.
- **Wrong email input type:** Using `type="text"` for email. Use `type="email"` so the browser validates format.
- **Label for attribute:** The **for** value in <label> must exactly match the input's **id**.

6. Summary of Your Two Practice Forms

Feature	Donation Form	Registration Form
Text input	✓ Full Name, Email	✓ First Name, Last Name
Password	✗ Not included	✓ Password field
Number input	✓ Phone, Amount	✗ Not included
Radio buttons	✓ Amount, Payment Mode	✓ Gender
Checkboxes	✓ Purpose	✓ Hobbies
Dropdown (select)	✓ Country	✓ Country
Textarea	✓ Comment	✓ Comment
File upload	✓ Payment Proof	✓ Profile Picture
required attr	✓ Used	✗ Not used
maxlength attr	✗ Not used	✓ Used
checked attr (default)	✗ Not used	✓ Gender default

■ Pro Tips to Remember:

- Always use `type="email"` for email fields — it provides built-in browser validation.
- Radio buttons sharing a group must have the **same** name but **different** ids.
- Use `required` on important fields so users can't skip them.
- The `placeholder` attribute is for hints only — it disappears when the user types.
- `name` = what the server receives. `id` = what the HTML/CSS/JS uses internally.

Keep practising — great work today! ■